ORIGINAL ARTICLE

Tislelizumab Plus Chemotherapy as First-Line Treatment for Locally Advanced or Metastatic Nonsquamous NSCLC (RATIONALE 304): A Randomized Phase 3 Trial

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ABSTRACT

Introduction: Tislelizumab, an anti–programmed cell death protein-1 antibody, was specifically engineered to minimize FcγR macrophage binding to abrogate antibody-dependent phagocytosis. Compared with chemotherapy alone, tislelizumab plus chemotherapy may improve clinical outcomes in patients with advanced nonsquamous NSCLC (nsq-NSCLC).

Methods: In this open-label phase 3 trial (RATIONALE 304; NCT03663205), patients with histologically confirmed stage IIIB or IV nsq-NSCLC were randomized (2:1) to receive either arm A: tislelizumab plus platinum (carboplatin or cisplatin) and pemetrexed every 3 weeks (Q3Ws) or arm B: platinum and pemetrexed alone Q3W during induction treatment, followed by intravenous maintenance

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Drs. Lu and J Wang were co-lead authors and contributed equally to this work.

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pemetrexed Q3W. The primary end point was progression-free survival (PFS) assessed by an independent review committee; clinical response and safety and tolerability were secondary end points.

Results: Overall, 332 patients (n = 222 [A]; n = 110 [B]) received treatment. With a median study follow-up of 9.8 months, PFS was significantly longer with tislelizumab plus chemotherapy compared with chemotherapy alone (median PFS: 9.7 versus 7.6 mo; hazard ratio = 0.645 [95% confidence interval: 0.462–0.902], $p = 0.0044$). In addition, response rates were higher and response duration was longer with combination therapy versus chemotherapy alone. Hematologic adverse events (AEs) were common in both treatment arms; the most reported AEs were grades 1 to 2 in severity. The most common grade greater than or equal to 3 AEs were associated with chemotherapy and included neutropenia (44.6% [A]; 35.5% [B]) and leukopenia (21.6% [A]; 14.5% [B]).

Conclusions: Addition of tislelizumab to chemotherapy resulted in significantly prolonged PFS, higher response rates, and longer response duration compared with chemotherapy alone, identifying a new potential option for first-line treatment of advanced nsq-NSCLC irrespective of disease stage.

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Keywords: Tislelizumab; Platinum-pemetrexed; Non-squamous non–small cell lung cancer; Stage IIIB; Stage IV

Introduction

Lung cancer is the leading cause of cancer death globally, with approximately 2.2 million new lung cancer cases and 1.8 million deaths in 2020.[1] Historically, platinum-based chemotherapy regimens have been the standard first-line therapy for patients with locally advanced or metastatic nonsquamous NSCLC (nsq-NSCLC) not harboring *EGFR* mutations or *ALK* rearrangements.[2] In recent years, a number of global phase 3 studies combining immune checkpoint inhibitors targeting programmed cell death protein-1 (PD-1) or programmed death-ligand 1 (PD-L1) with a platinum-based chemotherapy backbone in the first-line setting have revealed significantly improved survival benefits compared with chemotherapy alone.[3,4] Despite advances in treatment, the prognosis for patients with advanced nsq-NSCLC remains poor, leaving considerable room for improvement in patient outcomes.[5]

Tislelizumab is a humanized monoclonal antibody with high affinity and specificity for PD-1 that was specifically engineered to minimize FcγR binding on macrophages to abrogate antibody-dependent phagocytosis, a potential mechanism of T-cell clearance and resistance to anti–PD-1 therapy.[6,7] Reports from two early phase studies (BGB-A317-001 [NCT02407990] and BGB-A317-102 [NCT04068519]) revealed that single-agent tislelizumab was generally well tolerated and had antitumor activity in patients with advanced refractory solid tumors, including both squamous and nsq-NSCLC.[8,9] Tislelizumab in combination with platinum-based chemotherapy as first-line treatment for advanced SCLC and NSCLC, including nsq-NSCLC, resulted in robust responses in a phase 2 study (BGB-A317-206 [NCT03432598]).[10] In a phase 3 study, addition of tislelizumab to chemotherapy resulted in significantly improved progression-free survival (PFS) compared with chemotherapy alone in patients with stage IIIB or IV squamous NSCLC (RATIONALE 307; BGB-A317-307 [NCT03594747]).[11] Here, we present the efficacy and safety data from a multicenter phase 3 study (RATIONALE 304; BGB-A317-304 [NCT03663205]) comparing tislelizumab plus platinum-based chemotherapy versus platinum-based chemotherapy alone in patients with advanced nsq-NSCLC.

Material and Methods

Patients

RATIONALE 304 (BGB-A317-304) is an open-label, randomized, multicenter phase 3 study (Supplementary Fig. 1) conducted at 47 study sites in the People's Republic of China. The study was performed according to the ethical principles of the Declaration of Helsinki, Good Clinical Practice guidelines, and the requirements of the public registration of clinical trials. Written informed consent was obtained before participation in the study. A copy of the redacted protocol and statistical analysis plan can be found in the Supplementary Data.

Adult patients (aged 18–75 y) who were treatment naive for histologically confirmed locally advanced (stage IIIB) or metastatic (stage IV) nsq-NSCLC, as classified by the seventh edition of the American Joint Committee on Cancer Cancer Staging Manual, and without known sensitizing *EGFR* mutations or *ALK* rearrangements by tissue-based analyses, were eligible. Patients were eligible for inclusion if they were not amenable to curative surgery or radiotherapy, had measurable disease (Response Evaluation Criteria in Solid Tumors version 1.1), and had an Eastern Cooperative Oncology Group performance score of less than or equal to 1. Additional inclusion and exclusion criteria are listed in the Supplementary Data.

Study Design and Treatment

Patients were randomized 2:1 to receive either tislelizumab in combination with chemotherapy (arm A) or chemotherapy alone (arm B) using an interactive response

technology system. Randomization was stratified according to tumor cell (TC) PD-L1 membrane expression (<1% versus 1%–49% versus ≥50%) and disease stage (IIIB versus IV); patients whose tissues were not assessable for PD-L1 expression were included in the less than 1% TC PD-L1 subgroup. Tumor sample PD-L1 assessments are described in the Supplementary Data.

Patients in arm A received tislelizumab 200 mg plus platinum-based chemotherapy (carboplatin area under the curve 5 or cisplatin 75 mg/m^2 in combination with pemetrexed 500 mg/m^2) once every 3 weeks intravenously for four to six cycles (at investigator's discretion) during induction treatment, followed by maintenance tislelizumab plus pemetrexed treatment. Patients in arm B received platinum-based chemotherapy (carboplatin area under the curve 5 or cisplatin 75 mg/m^2 in combination with pemetrexed 500 mg/m^2) every 3 weeks for four to six cycles (at investigator's discretion) during induction treatment, followed by maintenance pemetrexed treatment. All patients received premedication with folic acid, vitamin B12, and glucocorticoids according to local clinical practice or local guidelines for use of chemotherapy. The choice of platinum-based drug was determined by the investigator before the initiation of study treatments. The treatments were continued until radiographic progression as per the Response Evaluation Criteria in Solid Tumors version 1.1 or until clinical progression, unacceptable toxic effects, investigator decision, or withdrawal of consent by patient. If toxicity was definitively attributed to one study drug, that specific drug could be discontinued. Patients in arm B could cross over to receive tislelizumab monotherapy on independent review committee (IRC)-confirmed disease progression.

End Points

The primary end point of this study was the comparison of PFS as assessed by the IRC (PFS_{IRC}) between tislelizumab plus carboplatin or cisplatin in combination with pemetrexed (arm A) versus carboplatin or cisplatin in combination with pemetrexed alone (arm B) in all randomized patients (intent-to-treat [ITT]). Key secondary efficacy end points assessed in the ITT analysis set included PFS as assessed by the investigator, PFS_{IRC} according to PD-L1 expression, objective response rate (ORR) assessed by the IRC, duration of response as assessed by the IRC (DoR_{IRC}), and overall survival (OS). Tumor assessments were scheduled every 6 weeks for the first 6 months and then every 9 weeks through the remaining 6 months of year 1 and every 12 weeks thereafter regardless of dose delays to manage toxicities. Definitions for each efficacy end point are included in the Supplementary Data.

The incidence and severity of adverse events (AEs) were a secondary safety end point and assessed in the safety analysis set, which included patients receiving any dose of tislelizumab or chemotherapy. Safety and tolerability were assessed throughout the trial by monitoring AEs, defined and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0. Potential immune-mediated AEs were selected from a list of preferred terms regardless of whether the investigator attributed the event to a treatment or considered the event to be immune related.

Statistical Analyses

Sample size was determined by the number of PFS events required to reveal superiority of tislelizumab plus chemotherapy (arm A) to chemotherapy alone (arm B). Planned enrollment was 320 patients; as such, the trial would have 85% power to reveal an hazard ratio (HR) of 0.65 for disease progression or death at a one-sided α of 0.025.

Categorical variables were summarized by the number (percentage) of patients; continuous variables were reported using descriptive statistics. Time-to-event end points were estimated using Kaplan-Meier analysis. Between-group differences in PFS were assessed using the stratified log-rank test. HRs and associated 95% confidence intervals (CIs) were calculated using a stratified Cox proportional hazard model with Efron's method for handling tied events. The randomization stratification factors were applied to all stratified efficacy analyses. A post hoc PFS interaction analysis for treatment and PD-L1 expression subgroups was performed, using a Cox model to estimate HR. The covariates in the model included treatment and PD-L1 expression (<1%, 1%–49%, ≥50%); their interaction with disease stage (IIIB versus IV) was a stratification factor.

One protocol-specified interim analysis was to be conducted after approximately 153 events; prespecified efficacy boundaries were one-sided ($p = 0.0078$). On the basis of the observed number of events (n = 159), the multiplicity-adjusted, one-sided alpha was adjusted to 0.0092 as determined by the Lan-DeMets O'Brien-Fleming spending function. An Independent Data Monitoring Committee reviewed the results from the interim efficacy analysis, and these interim analysis data are presented here.

Results

Patient Disposition, Demographics, and Baseline Disease Characteristics

Between July 23, 2018, and July 31, 2019, a total of 334 eligible patients were randomized to either arm A (n = 223) or arm B (n = 111) (Fig. 1). As of data cutoff, 117 of 334 patients (35.0%) in the ITT analysis set

remained on assigned treatment (arm A, n = 97 [43.5%]; arm B, n = 20 [18.0%]), with a median study follow-up of 9.8 months (95% CI: 9.23–10.38).

Demographics and disease baseline characteristics were representative of the target patient population and were generally well balanced between treatment arms, although the percentage of patients aged less than or equal to 65 years was numerically higher in arm A than arm B (Table 1).

Treatment Exposure

Overall, the median duration of tislelizumab treatment was 31.6 weeks (range: 3.0–78.4), and the median number of tislelizumab treatment cycles was 10 (range: 1–25). Most patients (n = 135; 60.8%) received nine or more cycles of tislelizumab (9–12 cycles, n = 73 [32.9%]; >12 cycles, n = 62 [27.9%]); median relative dose intensity of tislelizumab was 94.9 (range: 57–103) in arm A (Supplementary Table 1).

Median number of carboplatin or cisplatin cycles was 4.0 in both arms A and B; median number of pemetrexed cycles was 9.0 and 6.5 cycles in arms A and B, respectively. Chemotherapy exposure was similar between arms A and B regarding median relative dose intensity (Supplementary Table 2).

Efficacy Analyses

Significantly longer PFS_{IRC} was observed in arm A compared with arm B (HR = 0.645 [95% CI: 0.462–0.902], p = 0.0044) (Fig. 2*A*); median PFS_{IRC} was 9.7 months (95% CI: 7.7–11.5) and 7.6 months (95% CI: 5.6–8.0) in arms A and B, respectively. The estimated proportion of patients who were alive and had progression free at 12 months was 31.3% (95% CI: 21.7–41.4) in arm A and 16.7% (95% CI: 6.8–30.5) in arm B. PFS results were similar when evaluated by investigator's assessment (HR = 0.561 [95% CI: 0.411–0.767], p = 0.0001) (Supplementary Fig. 2).

In most subgroups evaluated, the observed PFS benefit was maintained with tislelizumab plus chemotherapy versus chemotherapy alone (Fig. 2*B*). In patients with stage IIIB disease, median PFS estimates were 9.0 (arm A) and 7.6 months (arm B); the risk of disease progression and death was reduced in the presence of tislelizumab plus chemotherapy compared with

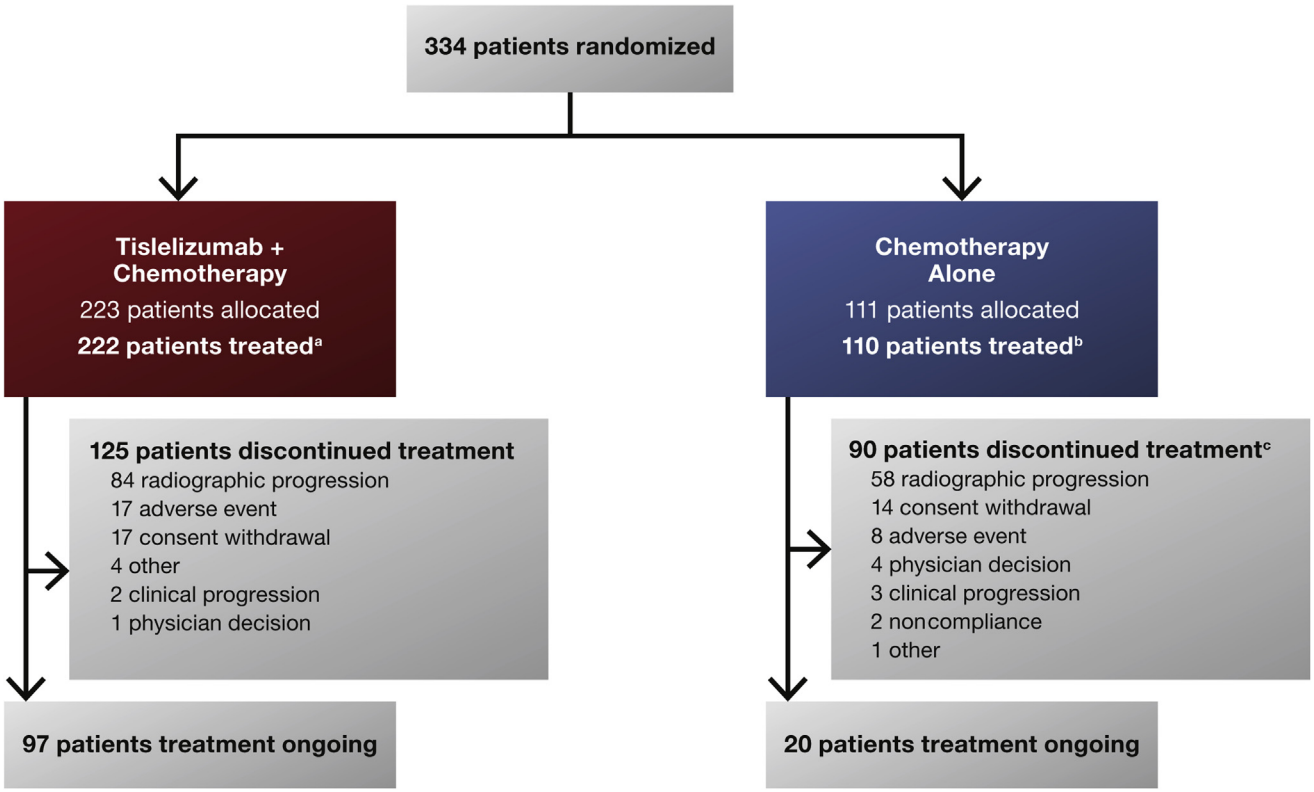


Figure 1. Patient disposition. Discontinued from treatment was defined as patients without any ongoing treatment drugs; primary reason for treatment discontinuation referred to primary reason for last discontinued study drug. [a]One patient randomized to combination therapy was not treated because inclusion criteria were not met. [b]One patient randomized to chemotherapy alone was not treated owing to withdrawal of consent. [c]A total of 26 patients crossed over to receive tislelizumab.

Table 1. Demographics and Baseline Characteristics of Patients With Advanced Nonsquamous NSCLC (ITT Analysis Set)

Characteristics	Arm A Tislelizumab + PP (n = 223)	Arm B PP (n = 111)	Total (N = 334)
Median age, y (range)	60 (27-75)	61 (25-74)	61 (25-75)
Age group, n (%)			
<65 y	163 (73.1)	74 (66.7)	237 (71.0)
≥65 y	60 (26.9)	37 (33.3)	97 (29.0)
Sex, n (%)			
Male	168 (75.3)	79 (71.2)	247 (74.0)
Female	55 (24.7)	32 (28.8)	87 (26.0)
Smoking status, n (%)			
Former	115 (51.6)	53 (47.7)	168 (50.3)
Current	32 (14.3)	13 (11.7)	45 (13.5)
Never	76 (34.1)	45 (40.5)	121 (36.2)
ECOG performance status, n (%)			
0	54 (24.2)	24 (21.6)	78 (23.4)
1	169 (75.8)	87 (78.4)	256 (76.6)
Disease stage at baseline, n (%)			
Stage IIIB	40 (17.9)	21 (18.9)	61 (18.3)
Stage IV	183 (82.1)	90 (81.1)	273 (81.7)
Histologic type, n (%)			
Adenocarcinoma	215 (96.4)	107 (96.4)	322 (96.4)
Mixed adenosquamous	1 (0.4)	2 (1.8)	3 (0.9)
Other	7 (3.1)	2 (1.8)	9 (2.7)
PD-L1% expression in tumor cells, n (%)			
<1[a]	96 (43.0)	48 (43.2)	144 (43.1)
1-49	53 (23.8)	27 (24.3)	80 (24.0)
≥50	74 (33.2)	36 (32.4)	110 (32.9)
EGFR sensitizing mutation status, n (%)			
Negative	218 (97.8)	109 (98.2)	327 (97.9)
Positive/unknown[b]	5 (2.2)	2 (1.8)	7 (2.1)
ALK rearrangement status, n (%)			
Negative	166 (74.4)	79 (71.2)	245 (73.4)
Unknown	57 (25.6)	32 (28.8)	89 (26.6)
Location of distant metastases, n (%)			
Bone	75 (33.6)	41 (36.9)	116 (34.7)
Liver	20 (9.0)	17 (15.3)	37 (11.1)
Brain	11 (4.9)	7 (6.3)	18 (5.4)
Any previous anticancer drug therapy, n (%)	15 (6.7)	8 (7.2)	23 (6.9)
Type of previous anticancer drug therapy, n (%)[c,d]			
Adjuvant	11 (73.3)	7 (87.5)	18 (78.3)
Neoadjuvant	2 (13.3)	0 (0.0)	2 (8.7)
Curative radiochemotherapy	1 (6.7)	0 (0.0)	1 (4.3)
Other	2 (13.3)	1 (12.5)	3 (13.0)

[a]Five patients with unassessable PD-L1 expression were included in PD-L1 less than 1% category.
[b]Included patients with *EGFR* sensitizing mutations or undeterminable status by means of tissue-based test, reported as major protocol deviations.
[c]Patients were counted once within each category but may have been counted in multiple categories.
[d]Percentages were based on the number of patients with any previous anticancer drug therapy.
ECOG, Eastern Cooperative Oncology Group; ITT, intent-to-treat; PD-L1, programmed death-ligand 1; PP, pemetrexed + platinum.

chemotherapy alone (HR = 0.664 [95% CI: 0.319–1.379]) (Supplementary Table 3). A similar trend was observed in patients with stage IV disease (9.7 versus 7.5 mo; HR = 0.632 [95% CI: 0.436–0.917]). In addition, PFS was prolonged with the addition of tislelizumab to chemotherapy versus chemotherapy alone in patients with negative (HR = 0.636 [95% CI: 0.434–0.930]) or unknown *ALK* rearrangement status (HR = 0.669 [95% CI: 0.345–1.297]) (Supplementary Table 4). PFS by TC PD-L1 expression (≥50%, 1%–49%, and <1%) are presented in Figures 2*C* to *E*. Post hoc exploratory interaction analyses suggested that TC PD-L1 expression of greater than or equal to 50% may be a possible predictor of PFS benefit, but these analyses were inconclusive for the other two PD-L1 expression groups (Supplementary Table 5).

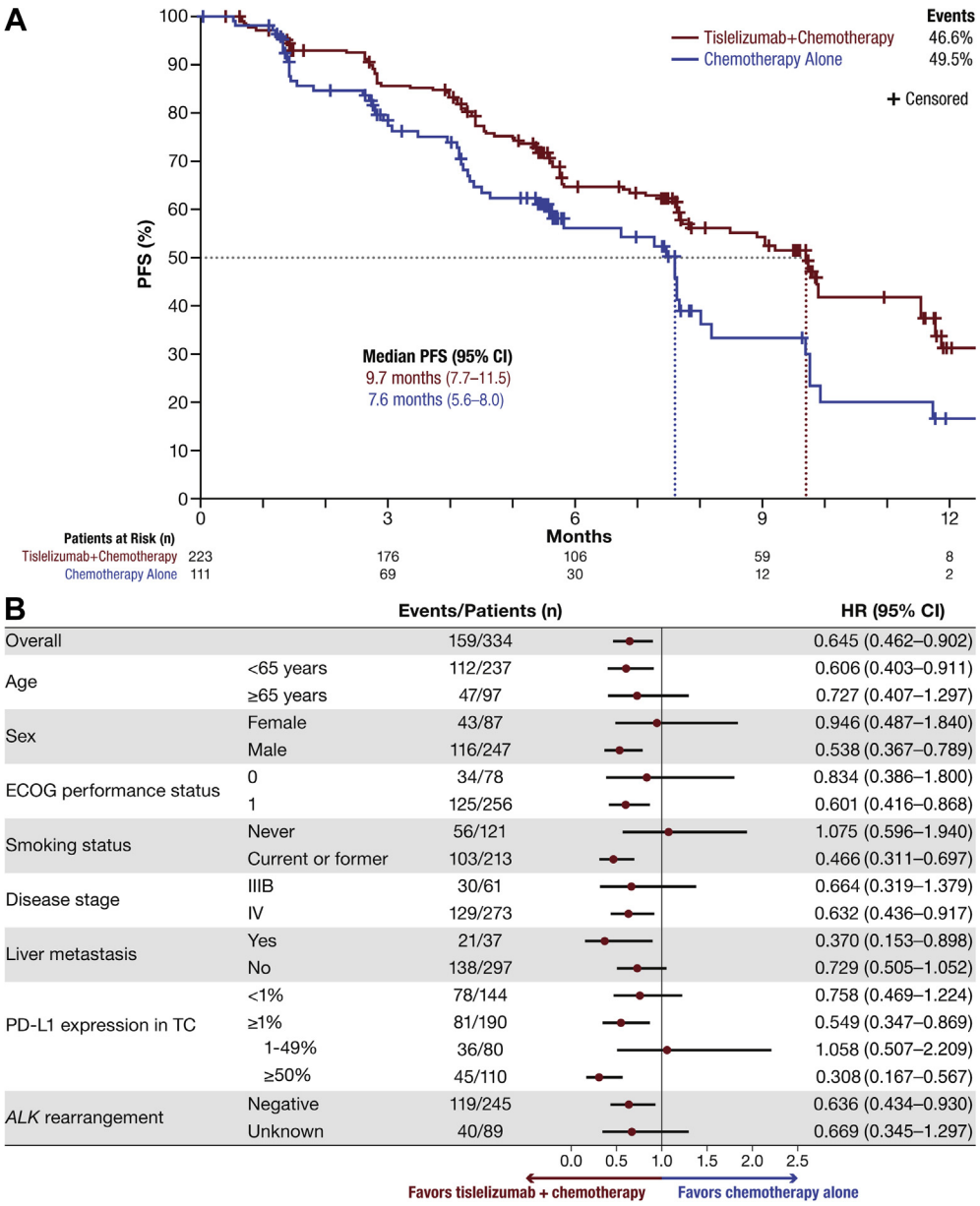


Figure 2. PFS by IRC (ITT analysis set, N = 334). (*A*) PFS (overall population). (*B*) PFS subgroup analysis. (*C*) PFS (≥50% TC PD-L1). (*D*) PFS (1%-49% TC PD-L1). (*E*) PFS (<1% TC PD-L1). CI, confidence interval; ECOG, Eastern Cooperative Oncology Group; HR, hazard ratio; IRC, independent review committee; ITT, intent-to-treat; NE, not estimable; PD-L1, programmed death-ligand 1; PFS, progression-free survival; TC, tumor cell.

A higher ORR assessed by the IRC was observed with tislelizumab plus chemotherapy (57.4%; 95% CI: 50.6–64.0) compared with chemotherapy alone (36.9%; 95% CI: 28.0–46.6) (Fig. 3). Among the 128 responders receiving tislelizumab combination therapy, median DoR_{IRC} was 8.5 months (95% CI: 6.80–10.58); in the 41 patients who achieved a response with chemotherapy alone, median DoR_{IRC} was 6.0 months (95% CI: 4.99–not estimable). Across all TC PD-L1 expression cutoffs, tislelizumab when combined with chemotherapy revealed higher ORR (Supplementary Fig. 3). With a median follow-up time of 9.8 months, median OS was not reached in either arm (Supplementary Fig. 4).

Safety and Tolerability

Overall, 222 patients (100%) in arm A and 109 patients (99.1%) in arm B experienced greater than or equal to one treatment-emergent AE (TEAE) (Supplementary Table 6). The most common TEAEs in both treatment arms were primarily hematologic in nature (e.g., anemia, leukopenia, thrombocytopenia) and

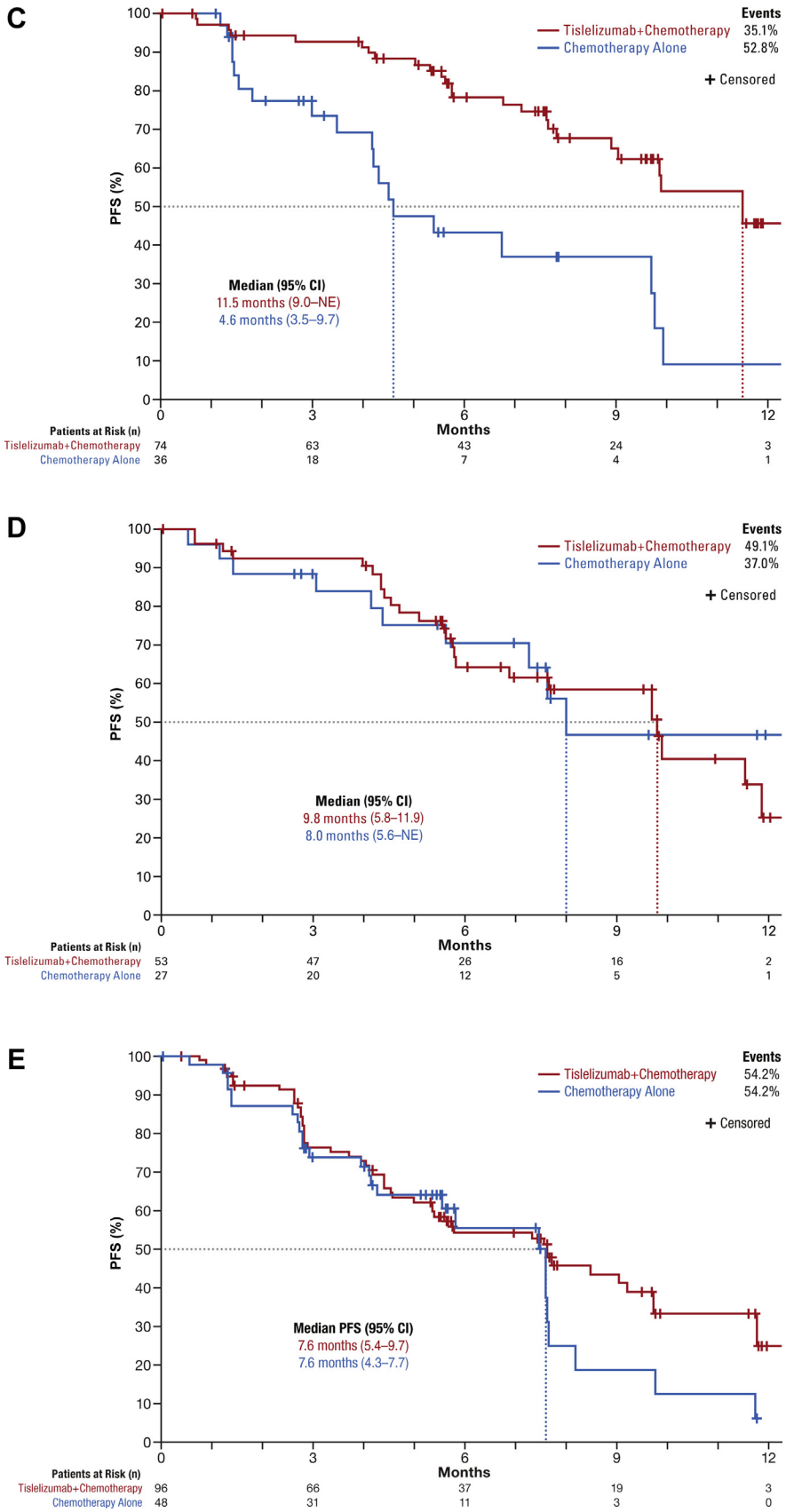


Figure 2. (*continued*).

most TEAEs were grades 1 to 2 in severity (Table 2). Serious TEAEs were reported in 97 patients (33.3% [A]; 20.9% [B]) (Supplementary Table 6). Discontinuation of any treatment component owing to TEAEs was reported in 25.7% (A) and 9.1% (B) of patients; TEAEs leading to permanent discontinuation of tislelizumab and dose modifications of tislelizumab occurred in 25 patients (11.3%) and 133 patients (59.9%), respectively.

AEs considered by the investigators to be related to any component of treatment were consistent with the overall TEAE profile (Supplementary Table 7). No differences were observed between the two arms in the most common grade greater than or equal to three treatment-related AEs (TRAEs) occurring in greater than or equal to 20% of the population; these grade greater than or equal to 3 AEs included neutropenia (44.6% [A]; 35.5%[B]), leukopenia (21.6% [A]; 14.5% [B]), thrombocytopenia (19.4% [A]; 13.6% [B]), and anemia (13.5% [A]; 10.0% [B]) (Supplementary Table 7). Serious TRAEs occurring in greater than or equal to 2% of patients included pneumonitis (5.9% [A]; 0.9% [B]), thrombocytopenia (3.2% [A]; 2.7% [B]), and decreased platelet count (2.3% [A]; 1.8% [B]). TRAEs leading to death occurred in three patients in arm A and one patient in arm B, respectively; all TRAEs leading to death were due to pneumonitis. Potential immune-mediated AEs were reported in 57 patients (25.7%) in the tislelizumab plus chemotherapy treatment arm. Pneumonitis (n = 20, 9.0%) and hypothyroidism (n = 19, 8.6%) were the most common potential immune-mediated AEs in patients receiving tislelizumab combination therapy (Supplementary Fig. 5). Most potential immune-mediated AEs were grades 1 to 2 and did not lead to treatment discontinuation.

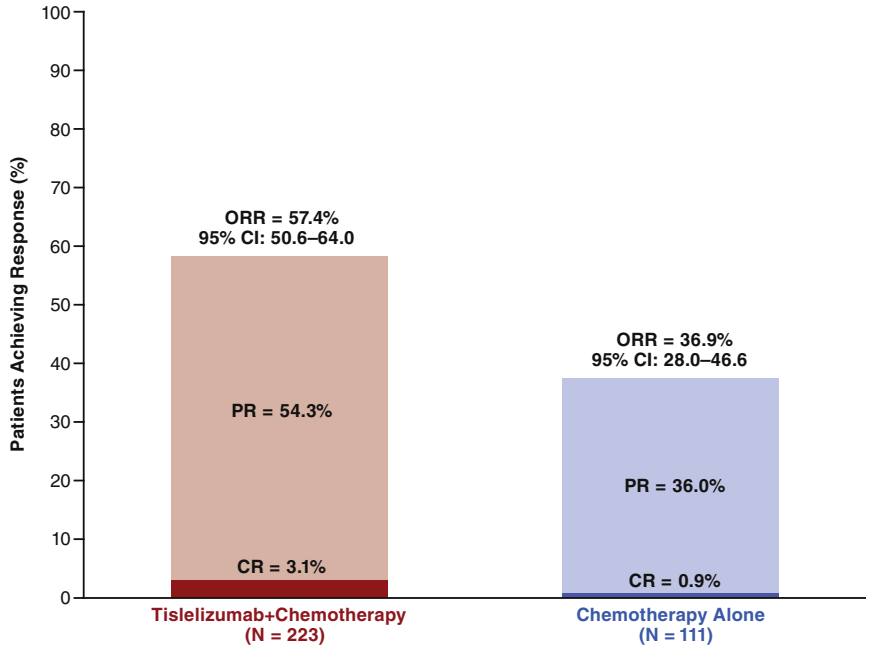
Discussion

The results of RATIONALE 304 revealed that the addition of tislelizumab to chemotherapy significantly prolonged PFS compared with chemotherapy alone as first-line treatment for locally advanced or metastatic nsq-NSCLC. In addition to significantly prolonged PFS, a higher proportion of patients achieved durable clinical responses with tislelizumab plus chemotherapy versus chemotherapy alone.

Although anti–PD-1 inhibitor monotherapy has revealed clinical benefit over chemotherapy alone as first-line treatment for patients with advanced NSCLC in patients with a total proportion score for PD-L1 expression of greater than or equal to 50%,[12] combining an anti–PD-1 or PD-L1 inhibitor with platinum chemotherapy revealed improved clinical outcomes in patients with advanced NSCLC regardless of PD-L1 expression.[3,4] In RATIONALE 304, tislelizumab revealed clinical benefit when combined with chemotherapy, driven by patients expressing high levels of TC PD-L1. In the 1% to 49% PD-L1 subgroup, PFS benefit was limited; this may be due to the small number of assessable patients and observed events. The greatest PFS benefit was observed in the subgroup of patients with greater than or equal to 50% PD-L1 expression, consistent with the results of previous studies of PD-1 pathway inhibition in advanced NSCLC.[4,13] A post hoc exploratory interaction analysis suggested that a PFS benefit was observed in patients with TC PD-L1 greater than or equal to 50%, but it was inconclusive for patients with either less than 1% or 1% to 49% TC PD-L1 expression levels.

The efficacy results of RATIONALE 304 are consistent with those from both KEYNOTE-189 and IMpower130[3,4]; however, unlike KEYNOTE-189 and IMpower130, RATIONALE 304 included patients with stage IIIB disease. Regardless of disease stage, the addition of tislelizumab to chemotherapy revealed prolonged PFS compared with chemotherapy alone, suggesting this treatment combination provides significant survival benefit to patients with either stage IIIB or stage IV disease. Moreover, prolonged PFS in patients with stage IIIB disease supports further evaluation of tislelizumab plus chemotherapy as first-line treatment for patients with locally advanced NSCLC that are ineligible for radiotherapy. RATIONALE 304 also identified consistent improvement of PFS duration with tislelizumab plus chemotherapy regardless of *ALK* rearrangement status, which may support inclusion of patients with unknown *ALK* rearrangement status at baseline. Nevertheless, this interpretation should be approached with caution, as some of the approximately 27% of patients with unknown *ALK* rearrangement status could have been *ALK*-positive, as the electronic case report forms did not capture the percentage of *ALK*-positive patients. It is not clear if the lower rate of brain metastases in RATIONALE 304 (5%), compared with what was reported in either KEYNOTE-189 or CheckMate 9LA (roughly 17% in both trials), influenced results of the trial.[3,14]

The value of checkpoint inhibitors in patients with *ALK*-rearranged NSCLC is not clear. Although IMpower130 revealed comparable outcomes with atezolizumab plus chemotherapy compared with chemotherapy alone in patients with known *ALK* rearrangement status, in IMpower150, atezolizumab combined with chemotherapy resulted in longer PFS in patients with *EGFR* or *ALK* alterations versus chemotherapy alone.[4,15] In addition, pemetrexed included in the chemotherapy regimen may have contributed to these findings, as pemetrexed has revealed benefits in patients regardless of *ALK* rearrangement status. It has also been suggested that PFS on first-line platinum



	Tislelizumab+Chemotherapy (N = 223)	Chemotherapy Alone (N = 111)
Disease Control Rate, (95% CI)	89.2% (84.4–93.0)	81.1% (72.5–87.9)
Duration of Response, median (95% CI)	8.5 months (6.80–10.58)	6.0 months (4.99–NE)

Figure 3. Best overall response by IRC (ITT analysis set, N = 334). CI, confidence interval; CR, complete response; IRC, independent review committee; ITT, intent-to-treat; NE, not estimable; ORR, objective response rate; PR, partial response.

regimens containing pemetrexed may be prolonged in never or light smokers.[16] Although the nonsmoker subgroup did not reveal PFS benefits with combination treatment in this trial, these results are not atypical in this patient population.[17] As such, the results from this study help to define a patient population with nsq-NSCLC that may benefit from the combination of tislelizumab plus chemotherapy as first-line treatment.

Although improvement in OS is considered the most convincing measure of clinical benefit in oncology drug trials, it can be confounded by in-trial crossover, effective subsequent-line therapies, and mortality unrelated to disease.[18] RATIONALE 304 was therefore designed with PFS as the primary end point to provide the best treatment for study participants and reduce the impact of these confounders on the study results. In addition, the open-label aspect of RATIONALE 304 may have affected study results. Nevertheless, specific protocol-defined procedures were taken to ensure the integrity of the data, including an independent radiologic review of the primary end point and a review of the results by an Independent Data Monitoring Committee.

First-line tislelizumab in combination with chemotherapy was generally well tolerated. The incidence of AEs reported in arm A was consistent with the known tolerability profile of PD-1 inhibitors in combination with chemotherapy and was also consistent with previous reports of tislelizumab-combination therapy.[3,10,13,19] Most reported AEs were mild or moderate in severity and were in line with the known toxicity profile of chemotherapy in this patient population. It is well known that antibodies against PD-1 or PD-L1 can result in increased immune activation, leading to potential immune-mediated AEs, especially pneumonitis in patients with NSCLC.[20] In RATIONALE 304, pneumonitis was the most common grade greater than or equal to 3 immune-mediated AE. The frequency of total immune-mediated pneumonitis with addition of tislelizumab was less than 10%, and most of these AEs were mild to moderate in severity. Four patients experienced fatal pneumonitis throughout this study, which is in line with previous reports. In KEYNOTE-189, three patients (0.7%) treated with pembrolizumab plus chemotherapy and two patients (0.4%) treated with atezolizumab and chemotherapy in IMpower130 experienced fatal pneumonitis.[3,4]

RATIONALE 304 was conducted entirely in the People's Republic of China, and the percentage of nonsmokers was much higher than what is typical in clinical trials of Western patient populations. For these reasons, the findings from the current trial may not transfer to patient populations that are more heterogenous. Nevertheless, the homogeneous nature of the patient population in RATIONALE 304

Table 2. Incidence of TEAEs Occurring in Greater Than or Equal to 20% of Patients (Safety Analysis Set, N = 332)

	Arm A Tislelizumab + PP (n = 222)		Arm B PP (n = 110)	
Incidence of TEAEs Occurring in ≥20% of Patients	Grades 1-2	Grades ≥3	Grades 1-2	Grades ≥3
Anemia[a]	158 (71.2)	33 (14.9)	79 (71.8)	13 (11.8)
Leukopenia[b]	135 (60.8)	48 (21.6)	65 (59.1)	16 (14.5)
Thrombocytopenia[c]	112 (50.5)	43 (19.4)	55 (50.0)	15 (13.6)
Increased ALT	100 (45.0)	8 (3.6)	47 (42.7)	3 (2.7)
Nausea	96 (43.2)	1 (0.5)	44 (40.0)	1 (0.9)
Increased AST	91 (41.0)	5 (2.3)	51 (46.4)	0 (0.0)
Neutropenia[d]	83 (37.4)	99 (44.6)	42 (38.2)	39 (35.5)
Fatigue[e]	82 (36.9)	3 (1.4)	38 (34.5)	2 (1.8)
Decreased appetite	72 (32.4)	3 (1.4)	32 (29.1)	1 (0.9)
Vomiting	59 (26.6)	1 (0.5)	24 (21.8)	1 (0.9)
Musculoskeletal pain[f]	55 (24.8)	0 (0.0)	19 (17.3)	2 (1.8)
Constipation	53 (23.9)	0 (0.0)	24 (21.8)	0 (0.0)
Hypoproteinemia[g]	51 (23.0)	0 (0.0)	15 (13.6)	0 (0.0)
Rash[h]	46 (20.7)	1 (0.5)	13 (11.8)	0 (0.0)

Note: Data presented as n (%).
[a]Included reports of anemia, decreased hemoglobin, and decreased red blood cell count.
[b]Included reports of decreased white blood cell count and leukopenia.
[c]Included reports of decreased platelet count and thrombocytopenia.
[d]Included reports of decreased neutrophil count and neutropenia.
[e]Included reports of asthenia, fatigue, and malaise.
[f]Included reports of arthralgia, back pain, bone pain, flank pain, muscle spasms, musculoskeletal chest pain, musculoskeletal discomfort, musculoskeletal pain, musculoskeletal stiffness, myalgia, neck pain, and pain in extremity.
[g]Included reports of hypoalbuminemia and hypoproteinemia.
[h]Included reports of dermatitis allergic, drug eruption, eczema, erythema, rash, rash maculopapular, rash pruritic, rash pustular, and skin exfoliation.
ALT, alanine aminotransferase; AST, aspartate aminotransferase; PP, pemetrexed + platinum; TEAE, treatment-emergent adverse event.

suggests that Asian patients, particularly those expressing high levels of PD-L1, could benefit from treatment with tislelizumab.

In conclusion, RATIONALE 304 demonstrated that first-line tislelizumab in combination with chemotherapy resulted in significantly longer PFS with higher rates of response than chemotherapy alone in patients with locally advanced or metastatic nsq-NSCLC. Tislelizumab in combination with chemotherapy resulted in tolerable toxicities, and no new safety signals were identified.

CRediT Authorship Contribution Statement

Xiusong Qiu: Concept and design.

Shun Lu, Xinghao Ai, Yan Yu, Xinmin Yu, Yanping Hu, Zhiyong Ma, Xingya Li, Wu Zhuang, Yunpeng Liu, Weidong Li, Jiuwei Cui, Dong Wang, Wangjun Liao, Mengzhao Wang, Jianying Zhou, Zhehai Wang, Yuping Sun, Xiusong Qiu, Jie Gao, Yuanyuan Bao, Liang Liang, Jie Wang: Acquisition, analysis, or interpretation of data.

Shun Lu, Jie Wang, Jie Gao, Yuanyuan Bao: Initial development of the manuscript.

Shun Lu, Xinghao Ai, Yan Yu, Xinmin Yu, Yanping Hu, Zhiyong Ma, Xingya Li, Wu Zhuang, Yunpeng Liu, Weidong Li, Jiuwei Cui, Dong Wang, Wangjun Liao, Mengzhao Wang, Jianying Zhou, Zhehai Wang, Yuping Sun, Xiusong Qiu, Jie Gao, Yuanyuan Bao, Liang Liang, Jie Wang: Critical revision of the manuscript for important intellectual content.

Yuanyuan Bao: Statistical analysis.

Liang Liang: Administrative, technical, or material support.

Jie Gao, Yan Yu, Jiuwei Cui, Xinmin Yu, Yanping Hu, Zhiyong Ma, Xingya Li, Wu Zhuang, Yunpeng Liu, Weidong Li, Mengzhao Wang, Jianying Zhou, Yuping Sun: Supervision.

Data Sharing Statement

On request, and subject to certain criteria, conditions, and exceptions, BeiGene, Ltd., will provide access to individual de-identified participant data from BeiGene-sponsored global interventional clinical studies conducted for medicines (1) for indications that have been approved or (2) in programs that have been terminated. BeiGene will also consider requests for the protocol, data dictionary, and statistical

analysis plan. Data requests may be submitted to medicalinformation@beigene.com.

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Supplementary Data

Note: To access the supplementary material accompanying this article, visit the online version of the *Journal of Thoracic Oncology* at www.jto.org and at https://doi.org/10.1016/j.jtho.2021.05.005.

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